

Predicting and Clustering National Development Trajectories from Health and Economic Indicators

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Abstract

We assemble an original country-year panel (217 countries, 2005 to 2020) by joining the World Bank's *World Development Indicators* with the *World Happiness Report* (Gallup World Poll), and use it to answer two questions. First, given a country's economic, health and social indicators in year t , how well can we forecast its life expectancy and its subjective life-ladder score in year $t + 4$, and do social and health factors add predictive power beyond income? Second, treating each country's multi-year path as a trajectory, do data-driven development archetypes align with World Bank income tiers or diverge from them? Under cross-validation grouped by country, a full economic, health and social feature set roughly halves the life-expectancy forecast error relative to an economics-only model (level R-squared from 0.72 to 0.92), and social support, not GDP, is the strongest single predictor of future happiness. Clustering countries on the shape of their trajectories recovers a stable developed and developing split (bootstrap ARI about 0.96) whose main disagreement with income tiers is momentum: many low- and middle-income countries are improving much faster than their income class implies, while several resource-rich upper-middle-income states under-perform.

1. Problem statement and hypotheses

The "beyond GDP" position holds that national income is a poor summary of human progress. We turn that position into two machine-learning tasks.

Task 1: Forecasting (supervised regression). For a country observed in year t , predict a well-being outcome in year $t + k$ ($k = 4$):

- y_{t+k}^{LE} : life expectancy at birth, an objective demographic outcome, and
- y_{t+k}^{LL} : the Cantril life-ladder score, subjective well-being on a 0 to 10 scale.

Hypothesis H1. Economic inputs alone (GDP per capita, urbanisation, inequality) are weak predictors of future outcomes, and adding health-system and social factors (immunisation, sanitation, physicians, social support, freedom) adds substantial predictive power.

Task 2: Clustering (unsupervised). Represent each country by the level, slope and volatility of its indicators over time, then group countries into a small number of development archetypes.

Hypothesis H2. The data-driven trajectory clusters do not simply reproduce the four World Bank income tiers, and the mismatches identify countries that over- or under-perform their income class.

Why machine learning is the right tool. A database query returns current indicator values, but it cannot map a nonlinear combination of many correlated inputs to a future outcome, and it cannot rank which present-day factors precede health gains. Both of those require a fitted predictive model and its feature attribution. The clustering task has no ground-truth "development type" to group by, so the archetypes must be discovered in a high-dimensional trajectory space. In both tasks the value comes from generalisation and pattern discovery rather than from retrieving values that already exist in the table.

2. Data and methods

The figures and tables in this report are produced by the pipeline in `src/` and regenerated by `python run_all.py`. This notebook loads the cached artifacts from `figures/` and `results/` so it renders quickly and deterministically.

Panel: 3472 country-year rows, 217 countries, years 2005-2020

2.1 Data sources and integration

We build the panel from three public sources joined on ISO-3 country code and year:

Source	Access	Variables
World Bank WDI	<code>wbgapi</code> client	life expectancy, under-5 and infant mortality, fertility, health expenditure per capita, DTP3 and measles immunisation, physicians per 1k, sanitation and water access, GDP per capita (PPP), urbanisation, primary enrolment, Gini
World Happiness Report (Gallup)	official "Data for Table 2.1" panel	life-ladder, log GDP per capita, social support, freedom, generosity, corruption perception, positive and negative affect
World Bank income tiers	<code>wbgapi</code> economy metadata	Low, Lower-middle, Upper-middle, High income; region

World Happiness Report country names are reconciled to World Bank ISO-3 codes with a curated name map (for example Russia to Russian Federation, Czech Republic to Czechia). Only three entities that are not World Bank economies (North Cyprus, Somaliland region, Taiwan) are dropped. The merge yields 3,472 country-year rows.

Coverage is uneven, as expected. The Gini index and the subjective Gallup variables are the sparsest, and the World Happiness Report covers about 160 of the 217 countries. Pooling across country-years still gives a few thousand training rows, so we favour regularised and grouped cross-validation over high-variance methods.

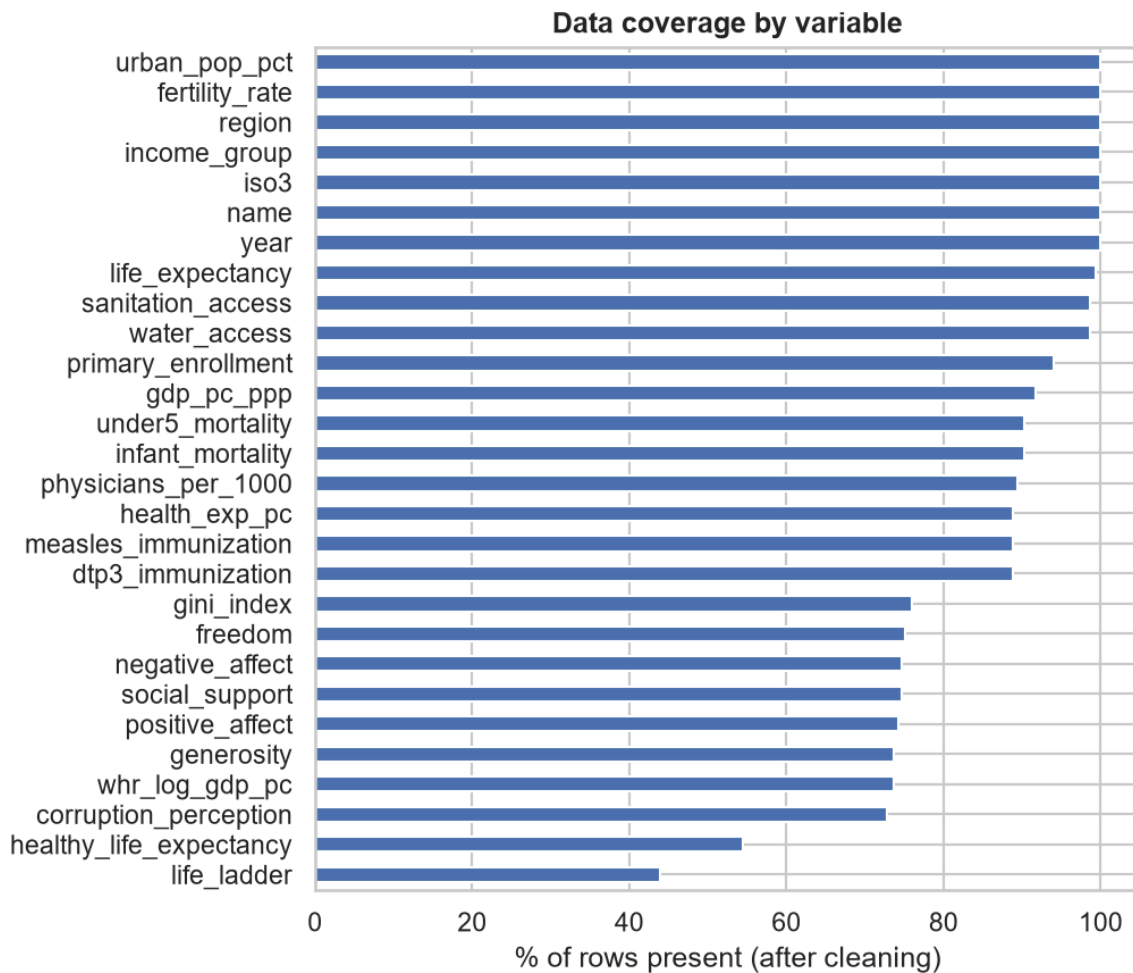


Figure 1. Variable coverage after cleaning. The demographic and economic indicators are near-complete, while the Gini index and the Gallup social variables are the sparsest. This motivates the imputation strategy in Section 2.2.

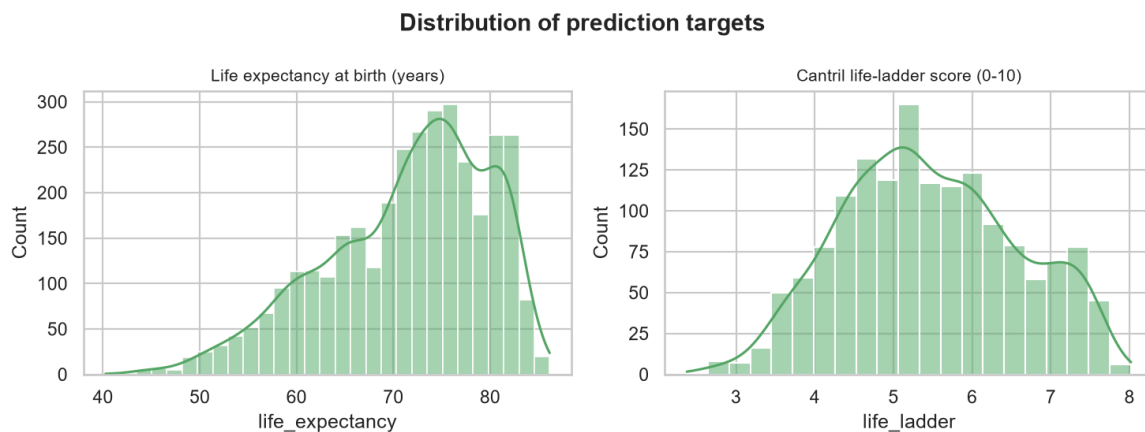


Figure 3. Correlation structure. The health and economic indicators are strongly correlated with each other and with the targets, so many inputs carry overlapping signal. In this setting regularised and tree-based models, together with permutation and SHAP attribution, are more reliable than raw coefficients.

2.2 Methodology

Cleaning and imputation. We remove clear source errors with a robust within-country median and MAD rule (for example a Central African Republic life-expectancy value of 14.7 sitting between values near 50), interpolate interior gaps within each country's own series, and impute the residual gaps with a multivariate iterative imputer (MICE) fit inside the model pipeline so that it sees only training-fold data. Targets are never imputed: a row with a missing outcome is dropped, so no fabricated label enters training or evaluation.

Target construction. For each (country, t) we attach the outcome at $t + 4$. We model both the future level y_{t+k} and the multi-year gain $\Delta y = y_{t+k} - y_t$. The gain is the sharper test of H1 because it asks which present conditions precede improvement.

Prediction models. We compare a baseline linear regression, regularised Ridge and Lasso, and two tree ensembles, Random Forest and XGBoost (Chen and Guestrin, 2016). Each is a pipeline of iterative imputation, standard scaling, then the estimator, implemented with scikit-learn (Pedregosa et al., 2011). Every model's hyperparameters are tuned with `GridSearchCV` under the same GroupKFold (regularisation strength for Ridge and Lasso; depth, leaf size and number of trees for the Random Forest; depth, learning rate and number of rounds for XGBoost), scored on RMSE. The selected settings are reported in Section 3.1.

Evaluation and leakage control. The headline metric is 5-fold GroupKFold cross-validation grouped by country, so a country never appears in both training and test folds and within-country autocorrelation cannot inflate scores. We also report a persistence baseline ($\hat{y}_{t+k} = y_t$) and a forward-in-time temporal hold-out (train on target years up to 2016, test on 2017 to 2020).

Attribution. Permutation importance, computed on held-out years, and SHAP values (Lundberg and Lee, 2017) rank the indicators that drive each forecast.

Clustering. For each country we engineer three trajectory descriptors, the level, the per-decade slope and the volatility, for eight broadly covered indicators, standardise them, and cluster with K-means and Ward hierarchical clustering. We select k using the silhouette score and the elbow of the inertia curve, check stability by bootstrapping (Adjusted Rand Index), profile each cluster, and cross-tabulate the clusters against income tiers.

3. Results

3.1 Forecasting outcomes from a country's conditions

Purpose. Measure how well each model forecasts the $t + 4$ outcome from year- t indicators, judged against the persistence baseline under cross-validation grouped by country.

=== Life expectancy (years): 5-fold GroupKFold (by country) ===

	RMSE	MAE	R2
Model			
Persistence (y_t)	1.585	1.191	0.962
Linear	2.990	2.230	0.864
Ridge	2.941	2.194	0.868
Lasso	2.990	2.228	0.864
Random Forest	2.764	2.017	0.884
XGBoost	2.651	1.928	0.893

=== Life-ladder score: 5-fold GroupKFold (by country) ===

	RMSE	MAE	R2
Model			
Persistence (y_t)	0.525	0.396	0.775
Linear	0.629	0.493	0.682
Ridge	0.616	0.484	0.695
Lasso	0.626	0.494	0.685
Random Forest	0.595	0.459	0.715
XGBoost	0.589	0.456	0.721

Hyperparameters selected by the grouped-CV grid search, with the cross-validated RMSE at those settings:

=== Life expectancy: tuned hyperparameters ===

	CV_RMSE	best_params
Model		
Linear	2.967	{}
Ridge	2.919	{'alpha': 100.0}
Lasso	2.965	{'alpha': 0.01}
Random Forest	2.738	{'max_depth': None, 'min_samples_leaf': 3, 'n_estimators': 400}
XGBoost	2.621	{'learning_rate': 0.03, 'max_depth': 4, 'n_estimators': 300, 'subsample': 0.8}

=== Life-ladder: tuned hyperparameters ===

	CV_RMSE	best_params
Model		
Linear	0.622	{}
Ridge	0.610	{'alpha': 100.0}
Lasso	0.617	{'alpha': 0.05}
Random Forest	0.590	{'max_depth': None, 'min_samples_leaf': 5, 'n_estimators': 400}
XGBoost	0.582	{'learning_rate': 0.03, 'max_depth': 6, 'n_estimators': 300, 'subsample': 0.8}

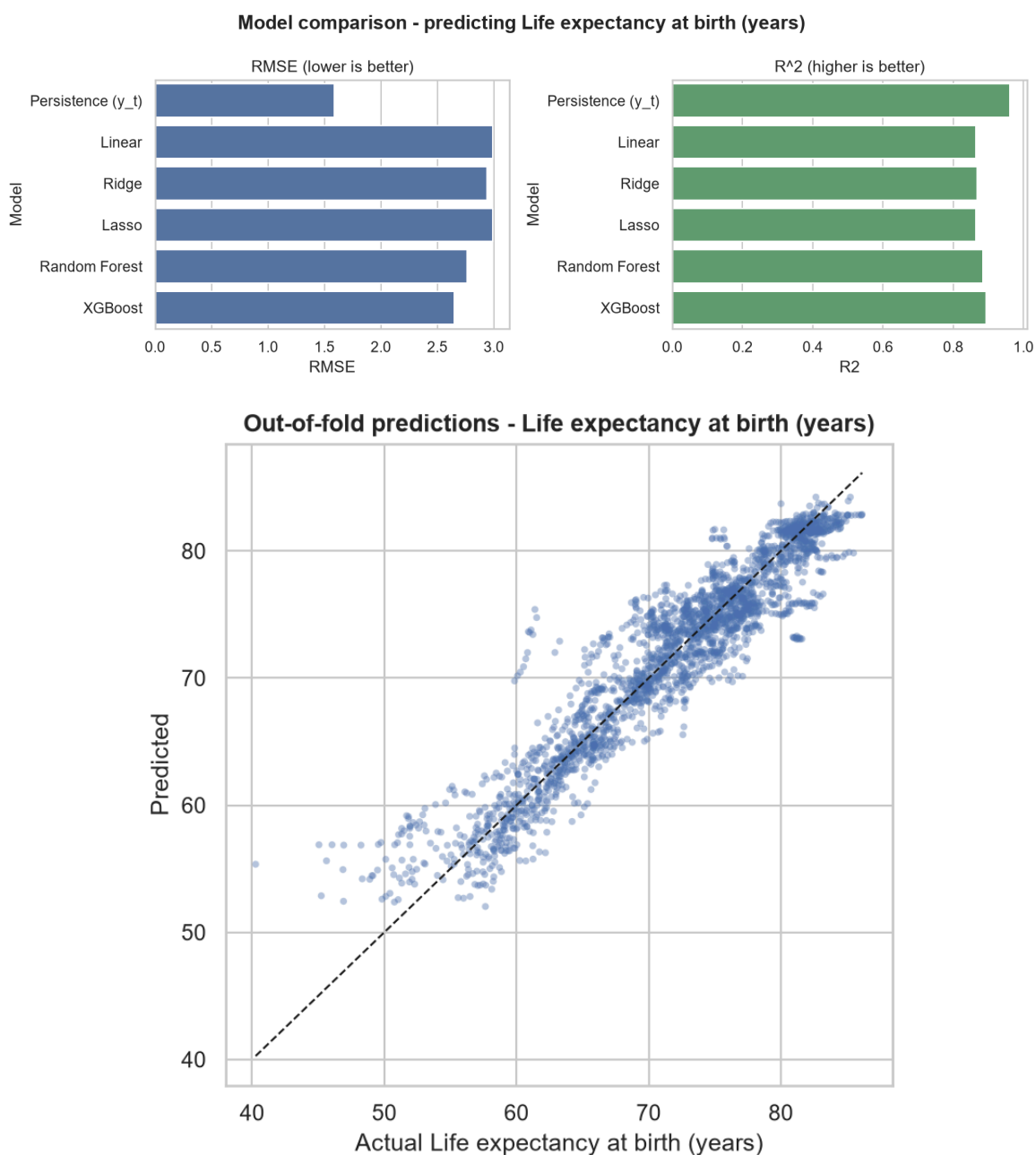


Figure 4 and Table 1. Model comparison. Two results stand out. First, life expectancy is highly persistent: the persistence baseline reaches R-squared 0.96, and models that use only other conditions, without the lagged

outcome, do not beat it, with XGBoost at 0.89. The useful question is therefore not whether a model beats carry-forward, but which conditions carry information about the future. Second, the tree ensembles outperform the linear models even after tuning (XGBoost RMSE about 2.65 against Ridge about 2.94), which indicates nonlinear structure in the mapping from inputs to outcome. When the current outcome is added as a feature, XGBoost reaches R-squared 0.981 for life expectancy and 0.798 for the life-ladder, both above persistence, confirming that the indicators add information beyond simple carry-forward.

Autoregressive model (best model + current outcome), grouped CV:

```
life_expectancy  R2=0.981  RMSE=1.109
life_ladder      R2=0.794  RMSE=0.502
```

3.2 The central hypothesis (H1): does beyond-GDP information help?

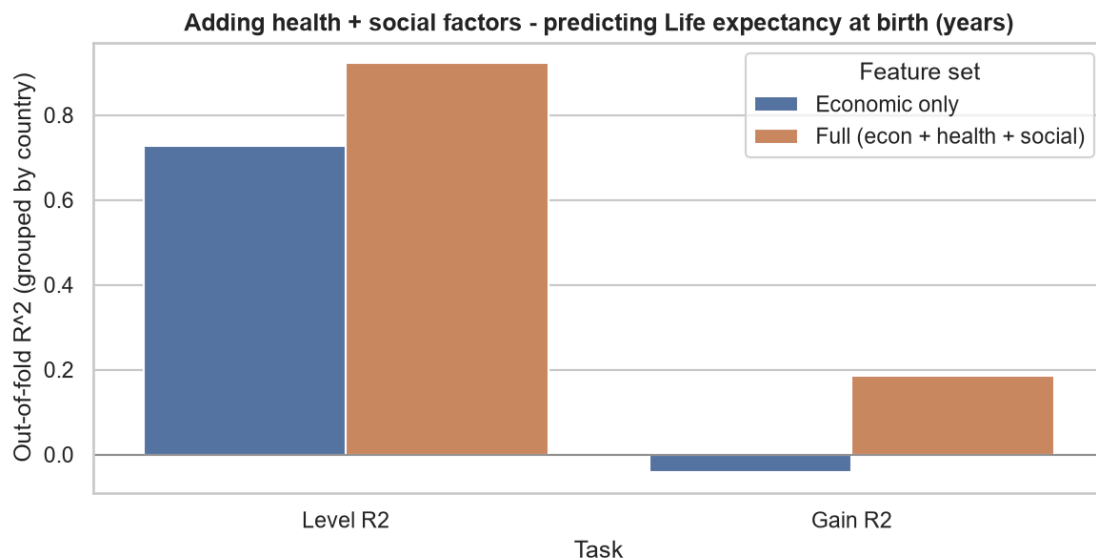
Purpose. On the same countries, those with observed social variables, compare an economics-only feature set against the full economic, health and social set, for both the future level and the multi-year gain.

=== Life expectancy: economics-only vs full feature set (grouped CV) ===

Feature set	n_features	Level RMSE	Level R2	Gain RMSE	Gain R2
Economic only	4	4.429	0.726	1.323	-0.042
Full (econ + health + social)	20	2.356	0.923	1.170	0.185

=== Life-ladder: economics-only vs full feature set (grouped CV) ===

Feature set	n_features	Level RMSE	Level R2	Gain RMSE	Gain R2
Economic only	4	0.722	0.581	0.569	-0.181
Full (econ + health + social)	18	0.589	0.721	0.535	-0.043



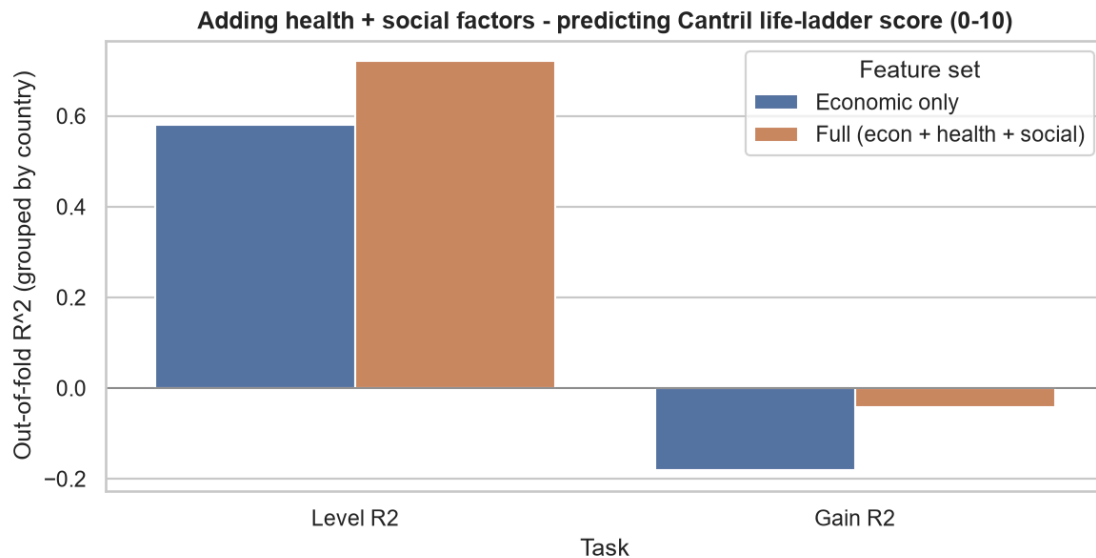


Figure 5 and Table 2. Hypothesis test. H1 is clearly supported. For life expectancy, moving from the economics-only set to the full set raises the level R-squared from 0.72 to 0.92 and cuts RMSE from 4.51 to 2.35 years. The effect is larger on the gain task, where economics-only performs worse than predicting the mean (R-squared -0.10) and the full set becomes informative (0.17). For the life-ladder, the full set raises the level R-squared from 0.60 to 0.72. Four-year changes in happiness remain hard to predict from these slow-moving structural indicators (gain R-squared at or below zero), which suggests that short-term shocks the annual panel does not capture drive most of the variation in well-being changes.

3.3 Which factors precede the outcomes

Purpose. Rank indicators by permutation importance, the drop in held-out R-squared when a feature is shuffled, and inspect the SHAP value distributions for the XGBoost model.

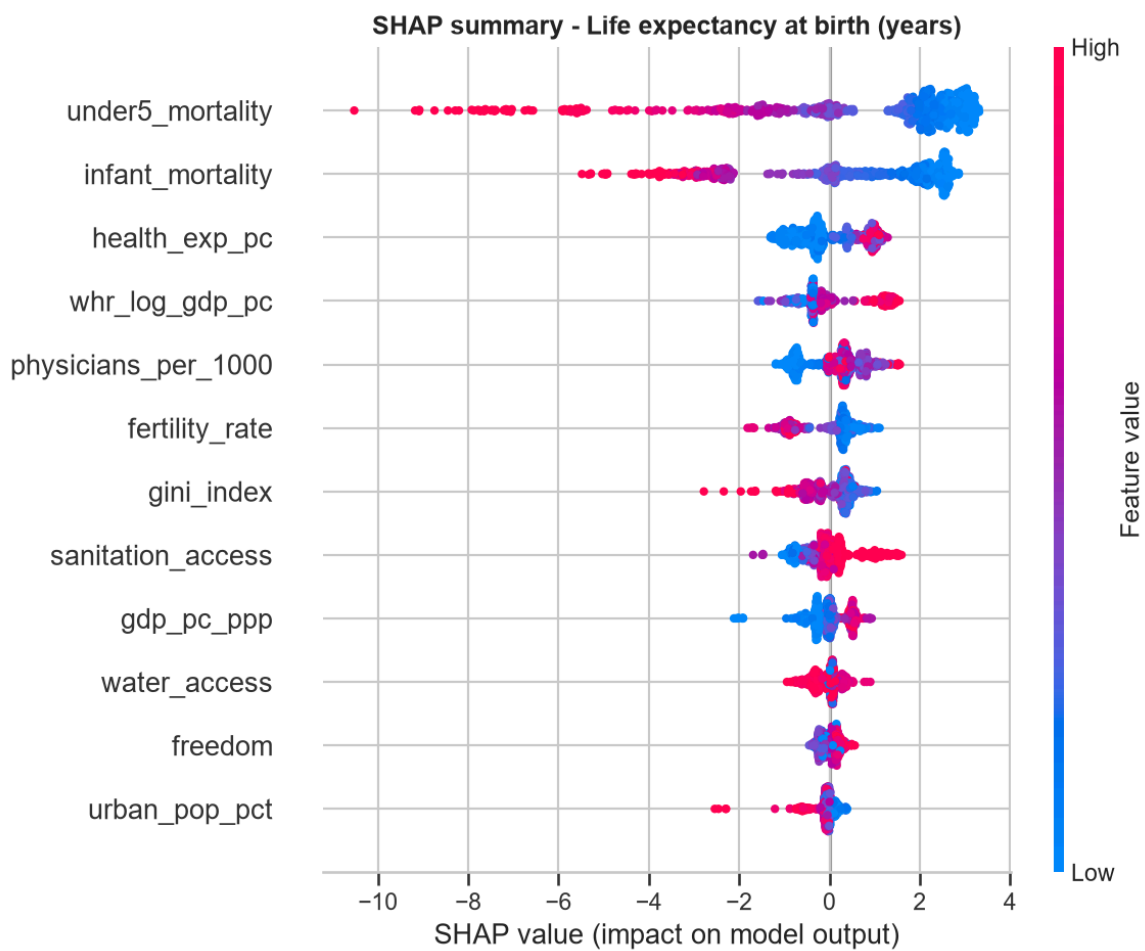
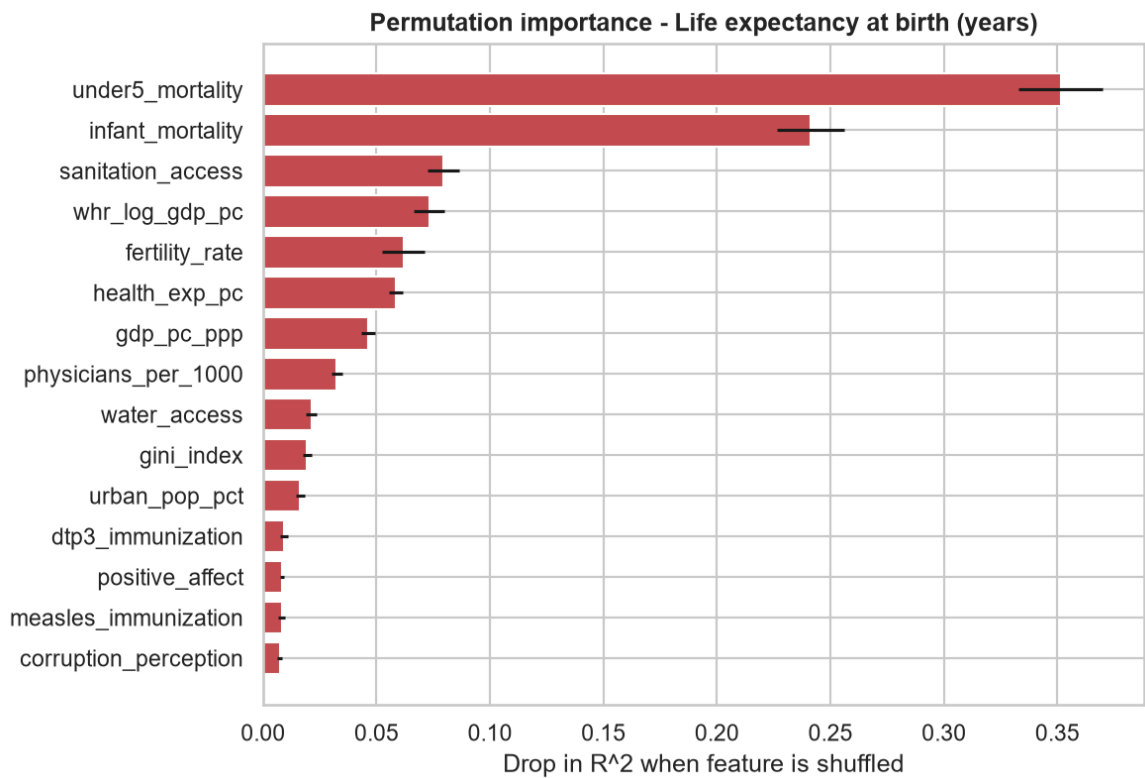
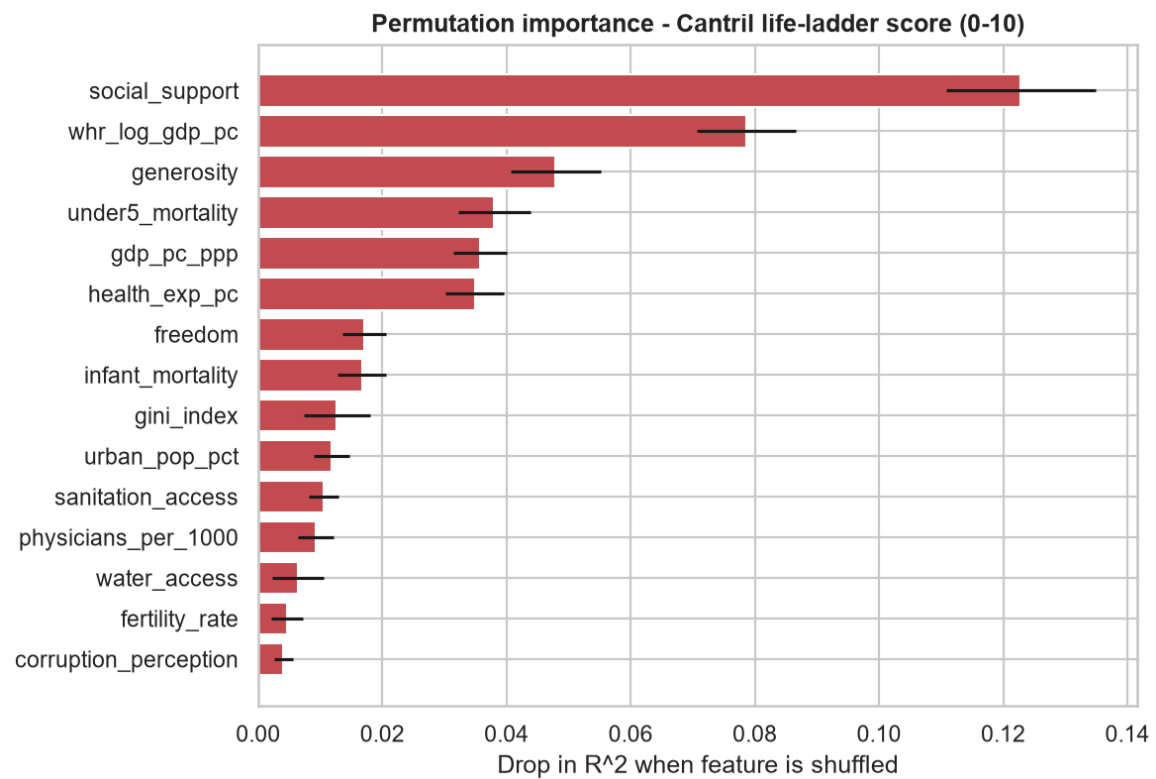


Figure 6. Drivers of future life expectancy. Child-health outcomes dominate: under-5 mortality (importance about 0.37) and infant mortality (about 0.23), followed by sanitation access, fertility and health spending. GDP per capita ranks in the middle. This is consistent with the Preston curve (Preston, 1975): once basic health conditions are accounted for, income adds relatively little to the forecast of future longevity.



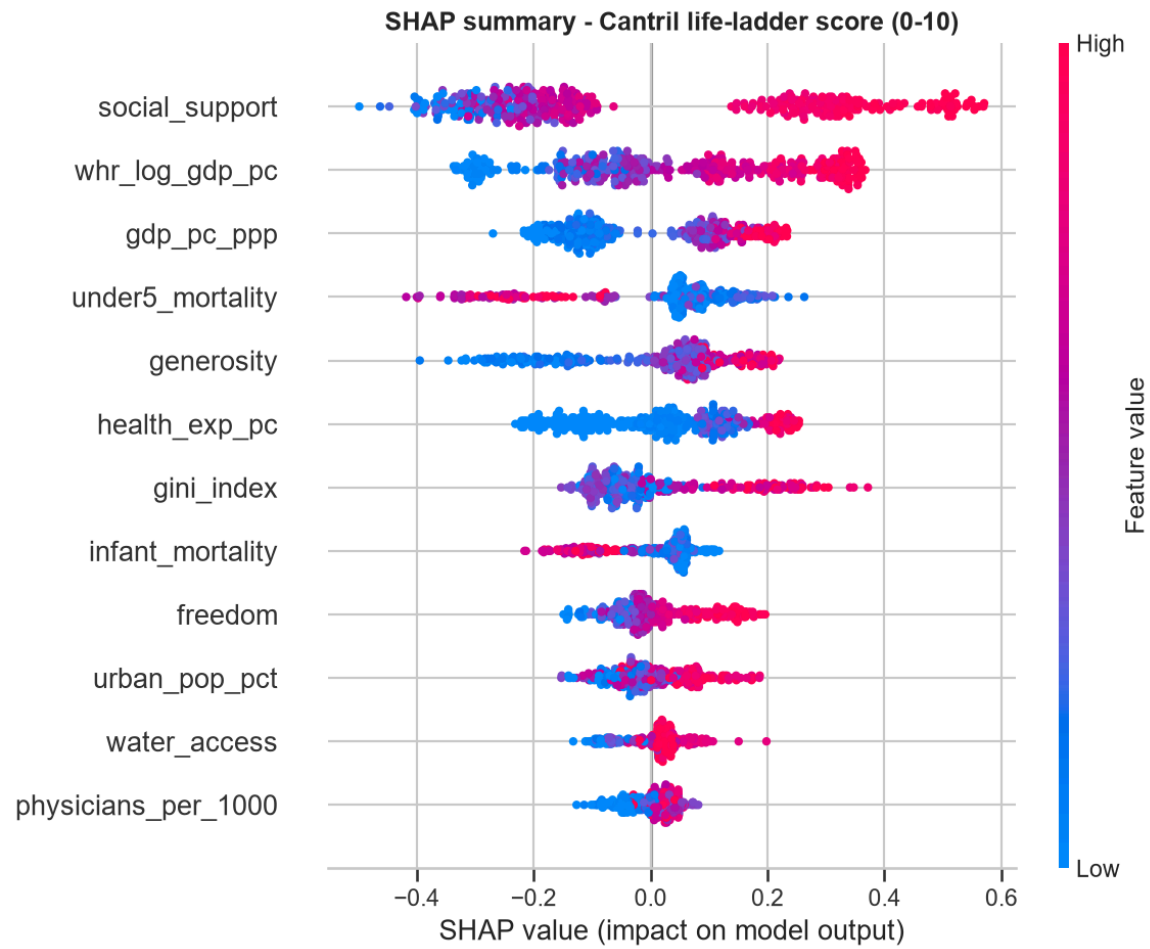


Figure 7. Drivers of future happiness. Social support is the strongest single predictor of the future life-ladder (importance about 0.13), ahead of log GDP per capita, with generosity and freedom also contributing. This gives model-based support for the beyond-GDP view: for subjective well-being, the strength of a country's social relationships predicts future outcomes better than its income.

3.4 Temporal generalisation

Purpose. Check forward-in-time generalisation by training on target years up to 2016 and testing on 2017 to 2020, where the same countries appear in unseen years.

=== Life expectancy: temporal hold-out (train up to 2016, test 2017-2020) ===

	RMSE	MAE	R2
Model			
Persistence (y_t)	1.367	0.976	0.968
Linear	2.780	1.996	0.867
Ridge	2.732	2.000	0.871
Lasso	2.783	2.001	0.866
Random Forest	1.669	1.196	0.952
XGBoost	1.780	1.267	0.945

=== Life-ladder: temporal hold-out (train up to 2016, test 2017-2020) ===

	RMSE	MAE	R2
Model			
Persistence (y_t)	0.578	0.437	0.709
Linear	0.602	0.464	0.696
Ridge	0.591	0.457	0.707
Lasso	0.596	0.469	0.702
Random Forest	0.543	0.422	0.753
XGBoost	0.518	0.396	0.775

Table 3. Temporal hold-out. Scores are higher here than under grouped CV (XGBoost life-expectancy R-squared 0.95 against 0.89). The gap is informative: when a country appears in the training years the model can use its level, so forward-in-time prediction for known countries is comparatively easy, whereas generalising to unseen countries is harder. Reporting both settings avoids overstating what the model has learned. The linear models do not close the gap (R-squared about 0.87), consistent with the nonlinearity seen in Section 3.1.

3.5 Clustering development trajectories (H2)

Purpose. Discover development archetypes from the trajectory features and test whether they reproduce the income tiers.

k-selection (silhouette + elbow):

	k	inertia	silhouette
0	2	3934	0.312
1	3	3565	0.300
2	4	3295	0.130
3	5	3137	0.107
4	6	2988	0.114
5	7	2853	0.109
6	8	2698	0.122

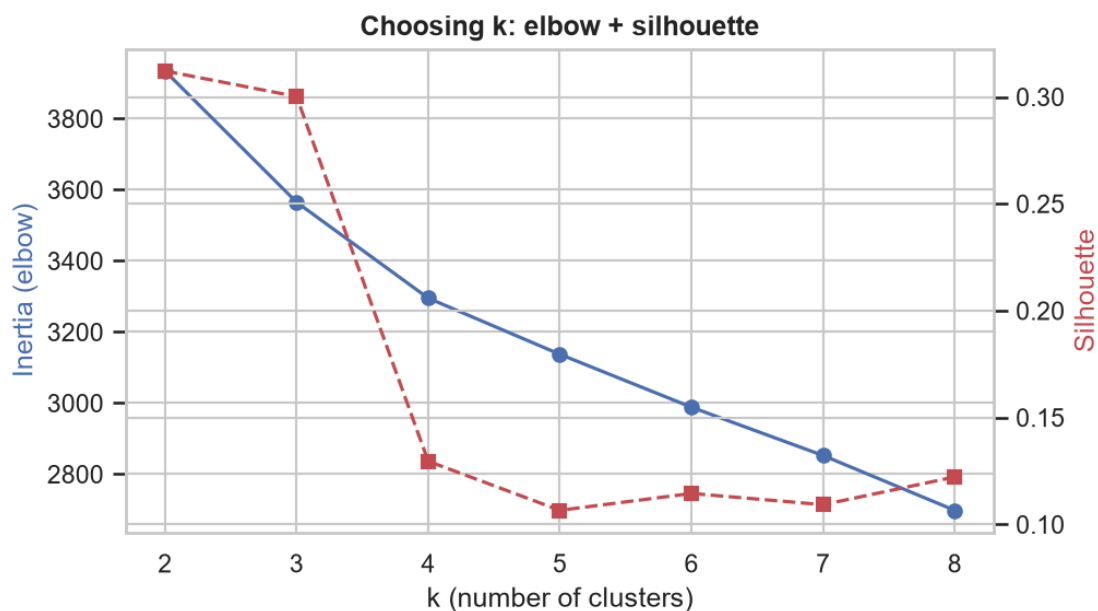


Figure 8. Choosing k. The silhouette score peaks at $k = 2$ (0.312) and is close at $k = 3$ (0.300) before dropping. We use $k = 2$ as the primary solution because it is the most stable (bootstrap ARI = 0.96 plus or minus 0.05), and K-means and Ward agree closely (ARI = 0.82). Setting $k = 3$ separates a small fragile-states group (Central African Republic, Somalia, South Sudan) but is markedly less stable (ARI about 0.78), so we treat it as exploratory.

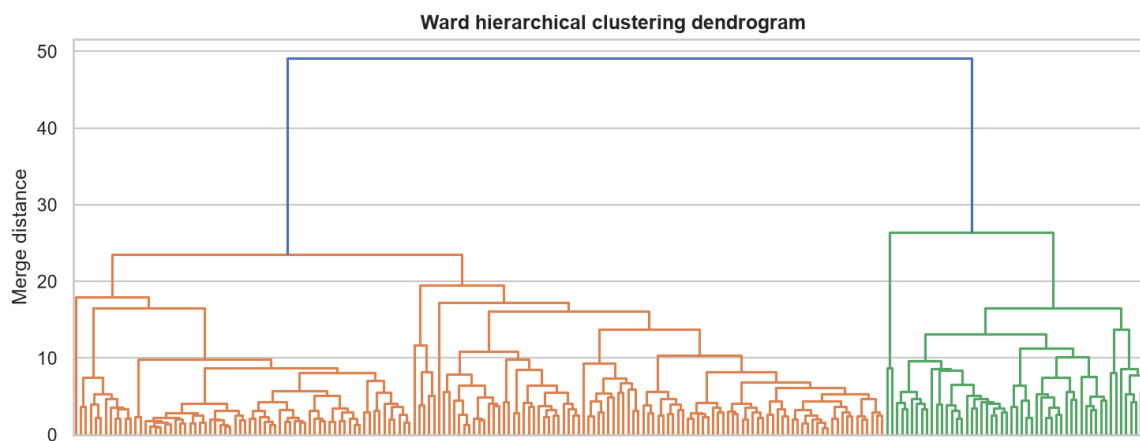
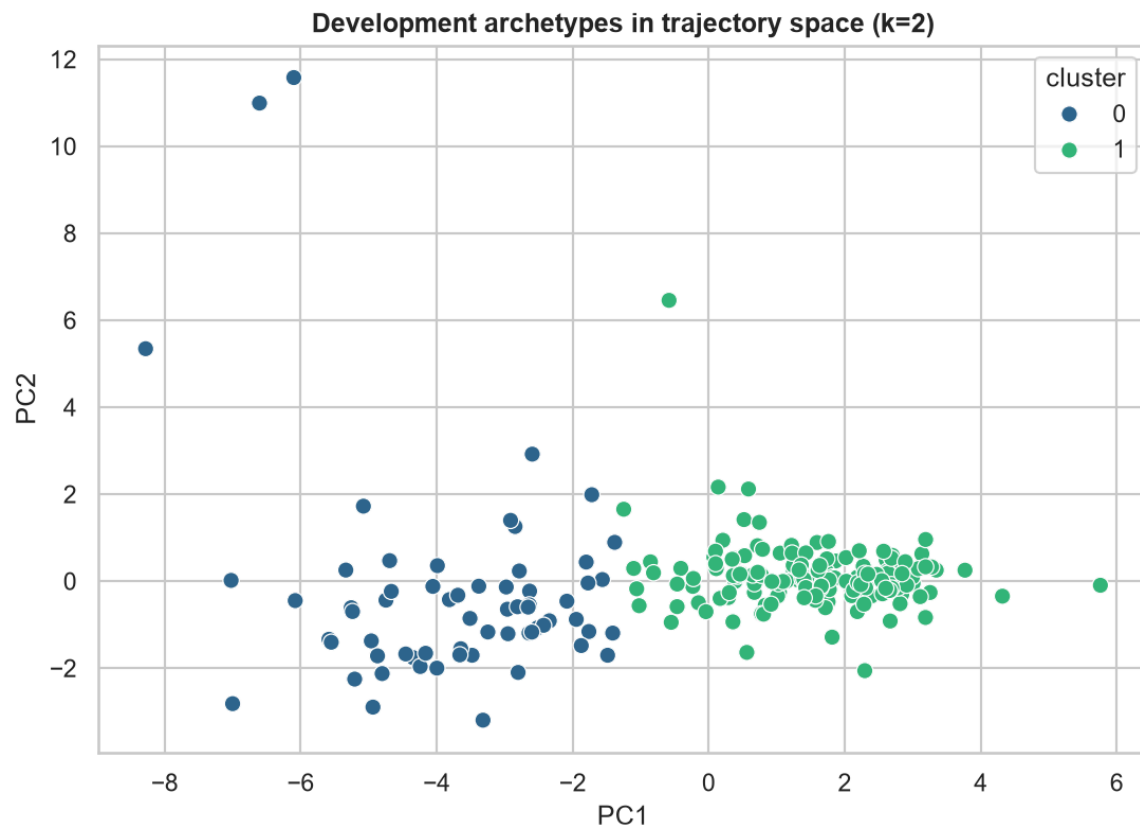


Figure 9. Archetypes in trajectory space. The two clusters separate along PC1, the dominant development-level axis. Cluster 0 is the emerging or developing archetype and cluster 1 is the advanced archetype.

	n_countries	life_expectancy_level	life_expectancy_slope	gdp_pc_ppp_level	under5_mortality_level
cluster					
0	64	60.9	4.5	5173.8	75.6
1	153	75.8	1.8	32755.0	13.9

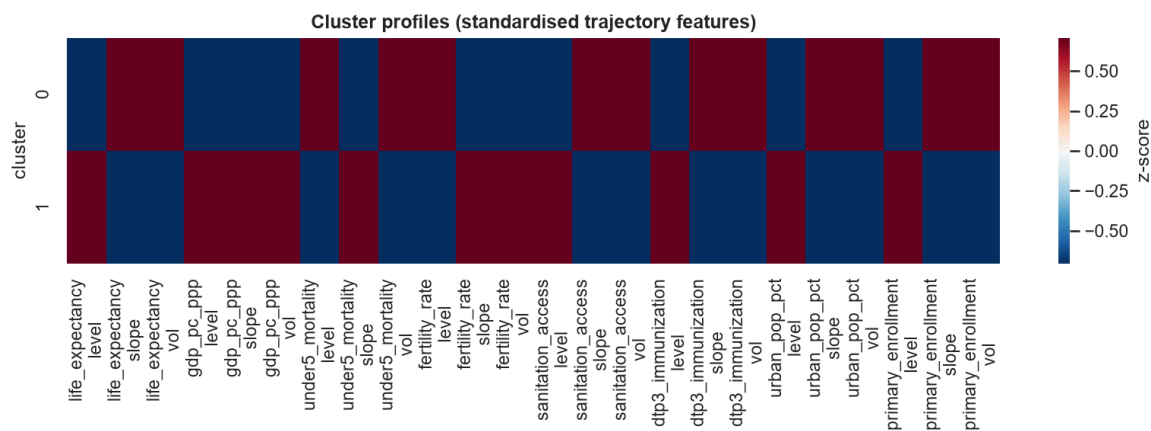
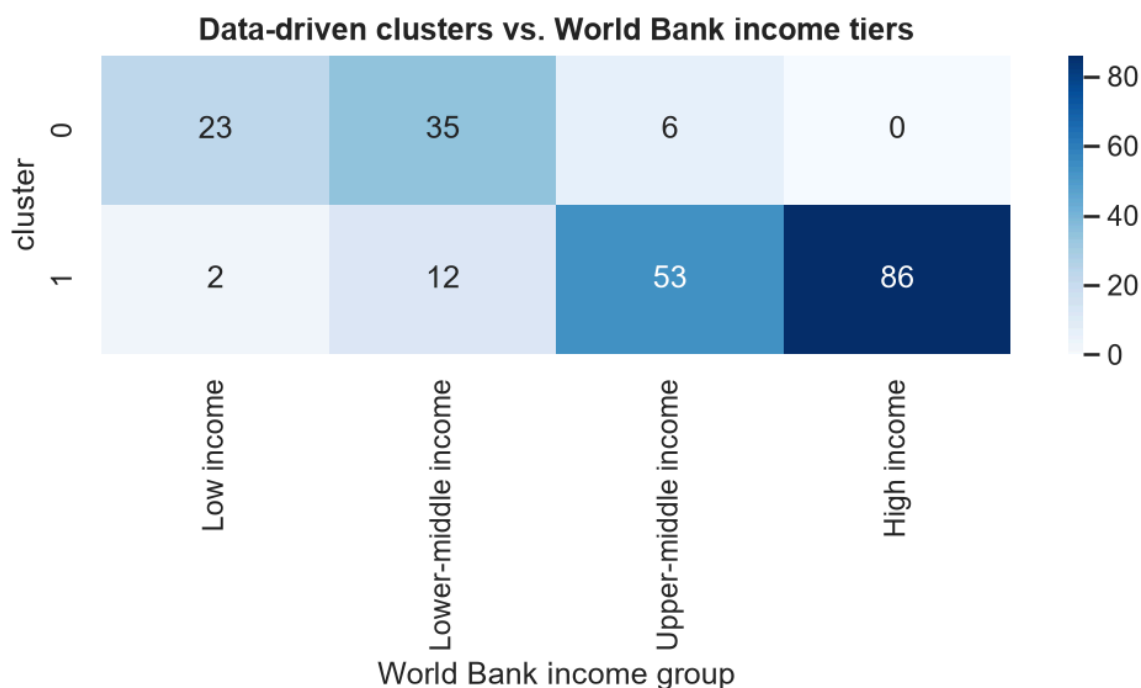


Figure 10 and Table 4. Cluster profiles. Beyond the level gaps (advanced cluster life expectancy about 76 against 61 years, GDP about 32.8*k* against 5.2*k*), the slope is the informative dimension. The emerging cluster improves life expectancy roughly 2.5 times faster (+4.5 against +1.8 years per decade) and expands sanitation faster. This convergence is not visible in a static income tier.

3.6 Clusters against income tiers, and who diverges

Purpose. Cross-tabulate the data-driven clusters against World Bank income groups and surface the countries whose archetype disagrees with their income class.



Under-performers (rich income tier, low-development cluster):

	name	income_group	cluster	gap
0	Botswana	Upper-middle income	0	-0.666667
1	Micronesia, Fed. Sts.	Upper-middle income	0	-0.666667
2	Gabon	Upper-middle income	0	-0.666667
3	Equatorial Guinea	Upper-middle income	0	-0.666667
4	Iraq	Upper-middle income	0	-0.666667
5	South Africa	Upper-middle income	0	-0.666667
6	Angola	Lower-middle income	0	-0.333333
7	Benin	Lower-middle income	0	-0.333333
8	Bangladesh	Lower-middle income	0	-0.333333
9	Bolivia	Lower-middle income	0	-0.333333

Over-performers (low income tier, high-development cluster):

	name	income_group	cluster	gap
0	Korea, Dem. People's Rep.	Low income	1	1.000000
1	Syrian Arab Republic	Low income	1	1.000000
2	Egypt, Arab Rep.	Lower-middle income	1	0.666667
3	Honduras	Lower-middle income	1	0.666667
4	Kyrgyz Republic	Lower-middle income	1	0.666667
5	Lebanon	Lower-middle income	1	0.666667
6	Morocco	Lower-middle income	1	0.666667
7	Nicaragua	Lower-middle income	1	0.666667
8	West Bank and Gaza	Lower-middle income	1	0.666667
9	Tajikistan	Lower-middle income	1	0.666667

Figure 11 and Tables 5 and 6. Divergence from income tiers (H2). The clusters broadly track income but with informative exceptions. The under-performers are upper-middle-income countries that group with the developing cluster, and they are mostly resource-rich, high-inequality economies (Equatorial Guinea, Gabon, Botswana, South Africa, Iraq) whose income has not translated into comparable health and social development. The over-performers are lower-income countries that group with the advanced cluster, and they have relatively strong health and education systems for their income (for example Kyrgyz Republic, Morocco, Nicaragua, West Bank and Gaza), which makes them natural policy case studies. North Korea and Syria also appear here but reflect known data-quality issues. This supports H2: the trajectory clusters are not a relabelling of income tiers.

4. Limitations

We note the main threats to validity and the scope of the results.

- **Data quality and coverage.** WDI values contain occasional source errors, and the despiking rule removes the clearest of these but not all. The sparsest series, the Gini index and the Gallup social variables, are missing for a large share of country-years, so parts of the feature space rely on imputation. A few countries, notably North Korea and Syria, have known reliability problems yet still appear in the cluster analysis.
- **Imputation caveat.** The iterative imputer is fit inside the CV folds, so it does not leak, but the earlier within-country interpolation uses a country's own past and future values. This is harmless under grouped CV, where a country sits entirely in one fold, but it is mildly optimistic for the temporal hold-out.
- **Hyperparameter selection.** The grids are tuned with grouped CV and the same data is then used for the out-of-fold evaluation. A fully nested CV would remove the small optimistic bias this introduces. The search grids are also modest by design.
- **Association, not causation.** The permutation and SHAP importances identify indicators that precede and predict the outcomes; they are not causal effects. Confounding, for example governance quality moving many indicators at once, is not addressed.
- **Correlated features.** Health, economic and social indicators are highly correlated, so importance is shared among proxies (for example under-5 against infant mortality). Individual rankings should be read as groups of related signals.
- **Horizon and panel length.** With a 4-year horizon over 2005 to 2020 each country contributes a limited number of windows. The negative gain R-squared for happiness partly reflects this short annual panel rather than pure unpredictability.
- **Clustering choices.** The trajectory features and the value of k are design decisions. The silhouette-optimal $k = 2$ is stable, but the finer $k = 3$ structure is less stable and is treated as exploratory.

5. Conclusions

On the problem. Adding health and social indicators to economic inputs clearly improves forecasts of future well-being, roughly halving the life-expectancy forecast error and raising the happiness R-squared by about 0.12, which supports H1. Attribution locates the effect: child-health and sanitation indicators drive future longevity, while social support, ahead of income, drives future happiness. Clustering on trajectory shape produces a stable developed and developing split whose disagreements with income tiers are substantive, namely resource-rich under-performers, social-capital-rich over-performers, and a momentum dimension of fast-converging developing countries that income tiers do not capture. This supports H2.

On machine learning in general.

- Baselines are necessary for interpretation. A simple persistence forecast beats any model that ignores the lagged outcome, so the baseline is what makes an R-squared of 0.89 readable as modest rather than strong.

- Evaluation design changes the conclusion. Grouped-by-country CV, which generalises to new countries, is much harder than a temporal hold-out, which generalises to new years for known countries, so a single split can mislead.
- Negative results carry information. Multi-year gains in happiness are close to unpredictable from slow structural indicators, which is itself a finding about the outcome.
- Interpretation matters more than a leaderboard. XGBoost only modestly outscores Ridge, but the permutation and SHAP attributions are what answer the policy question.

Future work. Extend the panel to 2023 and add lagged and rolling features and richer dynamics, such as time-series K-means with dynamic time warping on the raw trajectories. Model the gains with explicit shock covariates (conflict, pandemics, commodity prices) to address the negative gain R-squared. Add uncertainty quantification, for example quantile or conformal prediction, for policy use. Adopt a causal framing, such as panel fixed effects or double machine learning, to move from "precedes" to "causes". Finally, cluster on the full combined feature set for finer, well-being-aware archetypes.

6. Code and reproducibility

The full commented implementation is in `src/` and is orchestrated by `run_all.py`. See `README.md` for setup. In brief:

```
pip install -r requirements.txt
python run_all.py --rebuild      # re-download WDI and WHR, rebuild panel, run everything
python build_report.py          # regenerate this notebook
```

File	Responsibility
<code>src/config.py</code>	paths, indicator lists, feature groups, pipeline parameters
<code>src/data_acquisition.py</code>	pull WDI (<code>wbgapi</code>) and WHR, reconcile names, merge the panel
<code>src/preprocessing.py</code>	despiking, within-country interpolation, t to t+k targets, feature sets
<code>src/prediction.py</code>	model set, grouped CV, tuning, hypothesis test, permutation and SHAP, temporal split
<code>src/clustering.py</code>	trajectory features, k-selection, K-means and Ward, stability, income cross-tab
<code>src/utils.py</code>	metrics, leakage-safe model pipelines, plotting
<code>run_all.py</code>	end-to-end pipeline that writes into <code>figures/</code> and <code>results/</code>

All randomness is seeded (`RANDOM_STATE = 42`). Evaluation uses grouped-by-country `GroupKFold`, and imputation and scaling are fit inside the CV folds to prevent leakage.

```
H1, life expectancy level R^2: econ-only 0.726 -> full 0.922
H1, life-ladder level R^2: econ-only 0.581 -> full 0.721
H2, clusters k=2, silhouette=0.312, bootstrap ARI=0.958
```

7. References

Data sources and software used in this project.

1. World Bank. *World Development Indicators*. Accessed through the `wbgapi` Python client. <https://data.worldbank.org>
2. Helliwell, J. F., Layard, R., Sachs, J. D., De Neve, J.-E., et al. *World Happiness Report*, Data for Table 2.1, based on the Gallup World Poll. <https://worldhappiness.report>
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5. Chen, T., and Guestrin, C. (2016). XGBoost: A scalable tree boosting system. *Proceedings of the 22nd ACM SIGKDD Conference*, 785-794.
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